AP Precalculus | Unit 3, Lesson 3.10

Guided Notes

Name: ______________________________ Date: _________________ Period: _________

Primary Objective

I will be able to...

1. **LO 3.10.A** — Solve a trigonometric equation by isolating the trig function and using inverse trig to find all solutions. ______________________________
2. **LO 3.10.A** — Write the general solution of a trig equation using periodicity, and apply a domain restriction when context requires it. ______________________________

Vocabulary & Key Concepts

trigonometric equation: An equation in which the unknown appears inside a ________________________ function.

principal solution: The solution obtained by applying an _____________________ trig function directly; limited to the __________________ of that inverse function.

general solution: **All** solutions, expressed by adding integer multiples of the __________________ to the principal value(s). Example: $\theta = \frac{\pi}{6} + $ _________, $k \in \mathbb{Z}$.

domain restriction: A limit on the variable imposed by the ________________________ of the problem (e.g. $0 \le t \le 24$ for a 24-hour model).

LO 3.10.A — EK 3.10.A.2: Infinitely Many Solutions

Key Idea: Because trig functions are __________________, a trig equation typically has __________________ many solutions when the domain is all real numbers.

Example: Solve $\sin\theta = \dfrac{1}{2}$.

Step 1 — Principal value: $\theta = \arcsin\left(\frac{1}{2}\right) =$ ____________

Step 2 — Second solution in $[0, 2\pi)$: Sine is positive in Quadrants ______ and ______. Reflect: $\theta = \pi -$ _________ $=$ ____________

Step 3 — General solution: Add multiples of the period 2π to each solution.

Family 1: $\theta =$ __________________ Family 2: $\theta =$ __________________

where k is any __________________.

Check: List 3 specific solutions from each family:

Family 1 ($k = 0, 1, 2$): ______________________________

Family 2 ($k = 0, 1, -1$): ______________________________

LO 3.10.A — EK 3.10.A.1: The Four-Step Procedure

Algorithm for solving a trig equation:

1. **Isolate** the trig function on one side of the equation.
2. **Apply** the inverse trig function to find the principal value.

3. **Find** the second solution in $[0, 2\pi)$ using reference angles and the appropriate quadrant(s).
4. **Write** the general solution by adding $+2\pi k$ (or $+\pi k$ for tangent) to each solution.

Example: Solve $2\cos\theta - 1 = 0$.

Step 1 — Isolate: $\cos\theta =$ ____________

Step 2 — Principal value: $\theta = \arccos($______$) =$ __________

Step 3 — Cosine is positive in Q___ and Q___. Second solution: $\theta =$ ______________

Step 4 — General solution:
$\theta =$ ________________ or $\theta =$ ________________, $k \in \mathbb{Z}$

Solutions in $[0, 2\pi)$: ______________________________

LO 3.10.A — EK 3.10.A.3: Contextual Domain Restrictions

Context: The height of water at a harbor (in feet) is modeled by $h(t) = 3\sin\left(\frac{\pi}{6}t\right) + 5$, where t is hours after midnight, $0 \le t \le 24$.

Question: When is $h(t) \ge 7$? (When can a ship requiring 7 ft enter?)

Step 1 — Rewrite: $3\sin\left(\frac{\pi}{6}t\right) + 5 \ge 7 \quad \Rightarrow \quad \sin\left(\frac{\pi}{6}t\right) \ge$ __________

Step 2 — Let $u = \frac{\pi}{6}t$. Solve $\sin u = \frac{2}{3}$ for the boundary:

$u_1 = \arcsin\left(\frac{2}{3}\right) \approx$ __________ rad $\qquad u_2 = \pi - \arcsin\left(\frac{2}{3}\right) \approx$ __________ rad

Step 3 — Solve for t: $t = \dfrac{6}{\pi} \cdot u$

$t_1 \approx$ __________ hr $\qquad t_2 \approx$ __________ hr

Step 4 — The inequality $h(t) \ge 7$ holds on the interval ________ $\le t \le$ ________.

Interpret: The ship can enter between approximately ________ and ________ hours after midnight.

Why do we restrict to $[0, 24]$? ______________________________

Guided Practice

Guided Practice 1. Solve $\sqrt{2}\sin\theta - 1 = 0$ for all $\theta \in [0, 2\pi)$. Show the four steps. ________________

Guided Practice 2. Write the general solution to $\cos\theta = -\dfrac{\sqrt{3}}{2}$.

$\theta =$ ____________________ or $\theta =$ ____________________, $k \in \mathbb{Z}$

Guided Practice 3. A temperature model gives $T(t) = 10\sin\left(\frac{\pi}{12}t\right) + 65$ for $t \in [0, 24]$ (hours). For what hours is $T(t) \ge 70$?

Set up the inequality and isolate the trig function: ____________________

Solve for the boundary values of t: ____________________

Answer: $T(t) \ge 70$ on ____________________.